LEA Proteins During Water Stress: Not Just for Plants Anymore

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Abstract

Late embryogenesis abundant (LEA) proteins are extremely hydrophilic proteins that were first identified in land plants. Intracellular accumulation is tightly correlated with acquisition of desiccation tolerance, and data support their capacity to stabilize other proteins and membranes during drying, especially in the presence of sugars like trehalose. Exciting reports now show that LEA proteins are not restricted to plants; multiple forms are expressed in desiccation-tolerant animals from at least four phyla. We evaluate here the expression, subcellular localization, biochemical properties, and potential functions of LEA proteins in animal species during water stress. LEA proteins are intrinsically unstructured in aqueous solution, but surprisingly, many assume their native conformation during drying. They are targeted to multiple cellular locations, including mitochondria, and evidence supports that LEA proteins stabilize vitrified sugar glasses thought to be important in the dried state. More in vivo experimentation will be necessary to fully unravel the multiple functional properties of these macromolecules during water stress.

Compatible osmolyte: intracellular organic solute that helps maintain osmotic balance and cell volume and whose presence does not perturb macromolecular structure or function

Intrinsically disordered protein: a protein that lacks a well-defined, three-dimensional structure in aqueous solution

EST: expressed sequence tag

BLAST: basic local assignment search tool

POPP: protein or oligonucleotide probability profile

CS: citrate synthase

INTRODUCTION

Deficit in cellular water is a pervasive issue confronting both aquatic and terrestrial organisms. Subfreezing temperatures, desiccating conditions in xeric climates, and the osmotic variation seen in aqueous habitats are common environmental conditions that can impose severe water stress and a threat to life (1, 2). Freezing reduces the liquid water in extracellular fluids and can lead to intracellular dehydration in multicellular organisms. Many organisms facing water stress possess systems of compatible osmolytes, i.e., low-molecular-weight solutes accumulated in the intracellular compartment for osmotic balance. Other organic solutes like the sugar trehalose can stabilize biological structures during water stress (3–6). Small organic solutes are not the only components that contribute to an organism's desiccation tolerance, however. Protective macromolecules are also correlated with desiccation resistance and are of several types, including late embryogenesis abundant (LEA) proteins, small stress proteins like *Artemia* P26, Hsp 21 and Hsp 22 (7–16), and anhydrin (17). Both LEA proteins and anhydrin are examples of the extensive group of intrinsically disordered proteins (18, 19).

LEA proteins were first identified in land plants (20, 21), and their expression is associated with desiccation tolerance in seeds and anhydrobiotic plants (22). As the name suggests, these proteins were originally discovered in late stages of embryo development in plant seeds (20). Over the past several years, a series of remarkable findings have shown these proteins to be present in at least four animal phyla. In this review, we focus primarily on the expression, cellular localization, biochemical properties, and potential functions of LEA proteins in animal species. Broader reviews covering LEA proteins in nonanimal species are available (23–26).

DISTRIBUTION AND EXPRESSION OF LEA PROTEINS IN ANIMALS

LEA proteins are no longer viewed as being restricted to plants after having been documented in bacteria (27–29), cyanobacteria (30), slime molds (31), and fungi (32–34). In addition, reports of LEA-like protein in animals have steadily increased in the past several years. More than 30 protein sequences for LEA and LEA-like proteins (**Table 1**) appear in the National Center for Biotechnology Information (NCBI) database in animals such as nematodes (17, 35–40); rotifers (41–43); embryos of the brine shrimp *Artemia franciscana* (44–47); collembolan species, including the arctic springtail *Megaphorura arctica* (48, 49); the chironomid larva *Polypedilum vanderplanki* (50); and tardigrades [as referenced in an expressed sequence tag (EST) library (51)].

Relatedness among LEA proteins is demonstrated with bioinformatics tools such as the BLAST (basic local assignment search tool) algorithm, but its reliability can be partly compromised due to protein regions of low sequence complexity. Novel computational methods such as POPP (protein or oligonucleotide probability profile) analysis may offer additional insights into the relatedness of LEA protein sequences (52). Discovery of more LEA-related sequences in animals will also make their relatedness tractable to the traditional BLAST algorithm. New classifications of groups will most likely evolve with the development of new bioinformatics tools and the increase in available sequence data.

Two related but divergent LEA-like genes (*Arlea1A*, *Arlea1B*) located on two separate chromosomes are described in the bdelloid rotifer *Adineta ricciae* (53). Expression of both genes increases several-fold during desiccation. Interestingly, the protein sequences of both genes are similar, but their structure in the hydrated state differs substantially. ArLEA1A is largely unstructured in the hydrated state and assumes α-helical structure upon drying. ArLEA1B, in contrast, is predominately α-helical in the hydrated state. Furthermore, ArLEA1A prevents desiccation-induced aggregation of citrate synthase (CS), whereas ArLEA1B increases the aggregation of this enzyme and likely interacts preferably with phospholipid

Table 1 LEA-like proteins from animal species deduced from full-length nucleic acid sequences and selected expressed sequence tags (ESTs)

Species (identification)	Accession number	Length (number of amino acids)	Group	Hydropathy score	Reference(s)
Adineta ricciae (ArLEA1A)	ABU62809	421	LL (3)	−0.460	53
Adineta ricciae (ArLEA1B)	ABU62810	376	LL (3)	−0.465	53
Adineta vaga (LEA1B′)	ADD91479	354	LL (3)	−0.626	
Adineta vaga (LEA1C)	ADD91460	231	LL (3)	−0.694	
Aphelenchus avenae (AavLEA1)	AAL18843	143	3	−1.585	17, 35
Aphelenchus avenae (LEA-like 2)	ABQ23232	102	LL (3)	−1.376	
Aphelenchus avenae (LEA-like 1)	ABQ23233	85	LL (3)	−1.832	
Artemia franciscana (AfrLEA1)	ACA47267	357	3	−1.027	46
Artemia franciscana (AfrLEA2)	ACA47268	364	3	−0.884	46
Artemia franciscana (AfrLEA3m)	ACM16586	307	3	−1.295	44
Artemia franciscana	ACX81198[a]	217	1	−1.257	
Artemia franciscana (group 1 LEA)	ABX89317[a]	182	1	−1.410	45
Brachionus plicatilis (BpaLEA1)	ADE05593	613	LL (3)	−1.248	41, 42
Brachionus plicatilis (BpaLEA2)	ADE05594	248	LL (3)	−1.219	41, 42
Caenorhabditis briggsae (CbrLEA1)	XP_002637990	732	LL (3)	−1.104	
Caenorhabditis briggsae	XP_002638006	775	LL (3)	−0.899	
Caenorhabditis elegans (CeLEA1)	NP_001024042	733	LL (3)	−1.126	40
Caenorhabditis elegans (K08H10.2)	NP_505575	497	LL (3)	−1.054	
Ditylenchus africanus	FE923587	(142) EST	LL (3)	−1.422	39
Ditylenchus africanus	FE923426	(77) EST	LL (3)	−1.145	39
Megaphorura arctica[b]	EW755263	(147) EST	LL (3)	−1.297	48, 49
Polypedilum vanderplanki (PvLEA1)	BAE92616	742	3	−0.643	50
Polypedilum vanderplanki (PvLEA2)	BAE92617	180	3	−1.263	50
Polypedilum vanderplanki (PvLEA3)	BAE92618	484	3	−0.340	50
Pratylenchus penetrans	BQ627245	(87) EST	LL (3)	−1.383	
Steinernema carpocapsae	ABQ23231[c]	95	LL (3)	−1.194	37
Steinernema feltiae (SfLEA1)	AAM81356	102	LL (3)	−1.321	36, 38

[a]Both sequences are highly similar and differ nearly exclusively in the presence of a mitochondria-targeting sequence in the deduced protein for ACX81198.1. Other similar sequences are ABR67402.1, ADE45145.1, ADE45146.1, and ACX81197.1.
[b]Formerly *Onychiurus arcticus*.
[c]LEA-related proteins with similar sequences are reported for this species (ABQ23240.1 and ABQ23230.1).
LL denotes LEA like or LEA related.

membranes (53). The hydropathy score for both proteins is −0.46, which is moderately hydrophilic (e.g., bovine serum albumin = −0.43). LEA proteins are in general highly hydrophilic. The average hydropathy score for 30 group 3 LEA proteins from plants is −0.97 ± 0.3 (53, 54), and most LEA and LEA-like proteins from animals score less than −1 (**Table 1**). A homolog for ArLEA1B is reported for the closely related rotifer *Adineta vaga* (LEA1B′). This species also expresses a protein with a very similar sequence to LEA1B′ that has a deletion of 123 amino acids (LEA1C). The longer proteins ArLEA1A, ArLEA1B, and LEA1B′ have high sequence similarities to the dauer upregulated family member (Dur-1) protein from *Caenorhabditis elegans* (NP_001023145.1), with e-values of between $6e^{-8}$ and $1e^{-12}$.

Hydrophilic: water loving, i.e., the strong propensity of certain molecules to interact with water by hydrogen bonding

Anhydrobiosis: life without water, i.e., the ability to survive severe desiccation

Browne et al. (35) reported in 2002 the desiccation-induced expression of a group 3 LEA protein in the nematode *Aphelenchus avenae*. This protein is composed of 143 amino acids, is highly hydrophilic (with a hydropathy score of −1.585), and exhibits in its primary structure four copies of an 11-mer motif that is characteristic of group 3 LEA proteins (35). Sequence fragments for a different nematode species were reported earlier by Solomon et al. (36) and were annotated at that time as a LEA-like protein. Two other deduced sequences from *A. avenae* that are designated as LEA-like proteins appear in the NCBI database (**Table 1**).

At least five different LEA-like proteins are reported in the desiccation-tolerant embryo of the brine shrimp *A. franciscana* (44–46). The first three are designated as group 3 proteins (**Table 1**). AfrLEA1 is highly hydrophilic and exhibits sequence similarity with a *C. elegans* LEA-like protein (CeLEA1; $2e^{-12}$). AfrLEA2 exhibits homology to group 3 plant LEA proteins (e.g., *Brassica napus*, P13934.2, $2e^{-5}$). The mitochondria-localized protein AfrLEA3m shows high similarities to both the nematode protein CeLEA1 ($8e^{-7}$) as well as a LEA-like protein from the thale cress *Arabidopsis thaliana* ($8e^{-7}$). The two group 1 LEA proteins (45; **Table 1**) belong to the LEA_5 superfamily (pfam00477) and are the only representatives of LEA group 1 proteins in animals for which sequence data are available. These group 1 proteins from *A. franciscana* are virtually identical except for a mitochondria-targeting sequence and are highly similar to the EM1 protein from *A. thaliana* ($2e^{-25}$).

Figure 1 shows expression levels for the mRNA of the group 3 LEA proteins from *A. franciscana*. Each developmental stage with the capacity for anhydrobiosis (diapause and postdiapause embryos) expresses high message levels compared with the desiccation-intolerant larval stage (44, 46). Specifically, mRNA levels for *Afrlea1* and *Afrlea2* (which encode the two cytoplasmic LEA proteins) are 7-fold and 14-fold higher in the desiccation-tolerant stages compared with the intolerant stage, and similarly, mRNA for the mitochondrial *Afrlea3m* is elevated 9-fold and 11-fold (**Figure 1**). This differential expression is consistent with a role for these gene products in survivorship during dehydration. Similarly, Sharon et al. (45) reported expression of two mRNAs encoding group 1 LEA proteins in postdiapause embryos of *A. franciscana* but found no expression in larval stages. Chen et al. (47) generally confirmed these results.

Denekamp et al. (42) report, for the marine rotifer *Brachionus plicatilis*, three EST transcripts matching group 3 LEA proteins. Two LEA-like protein sequences (BpaLEA1 and BpaLEA2) are available in the protein database of NCBI (**Table 1**). Both proteins are highly hydrophilic but differ in size by approximately 360 amino acids. BpaLEA1 shows high similarities to CeLEA1 from *C. elegans* ($>8e^{-20}$) and to a group 3 LEA protein from the soybean *Glycine max*. BpaLEA2 exhibits high resemblance to a LEA domain–containing protein from *A. thaliana* (NP_193834.11, e^{-11}).

Multiple nematode species possess LEA proteins. Hydropathy values for these proteins

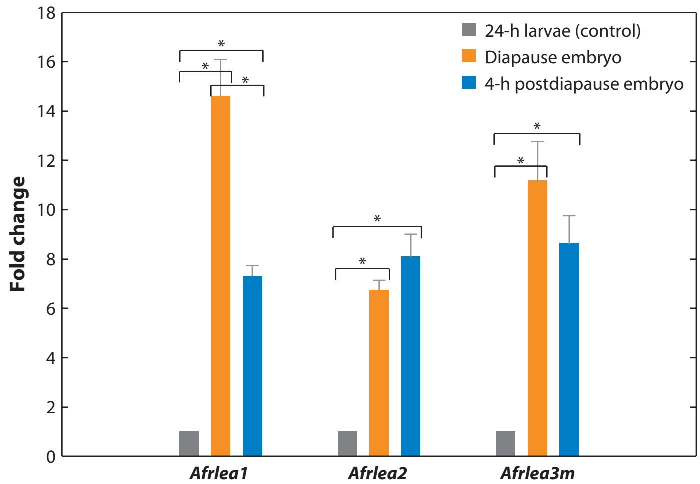

Figure 1

mRNA expression profiles for *Afrlea1, Afrlea2, and Afrlea3m* from *Artemia franciscana* embryos. LEA mRNAs are maintained at 7–14-fold-higher levels in the two desiccation-tolerant embryonic stages (i.e., diapause and postdiapause) compared with the desiccation-intolerant nauplius larvae that served as a control. Asterisks indicate that the paired means are statistically different (*t* test, $p < 0.05$; $N = 3$–5). Modified from References 44 and 46.

are generally less than −1. Two LEA-like proteins have been described in *C. elegans* (40) and the closely related species *Caenorhabditis briggsae* (**Table 1**). CeLEA1 RNA expression increases approximately 20-fold after 8 h of moderate desiccation stress in dauer juveniles (DJ) of *C. elegans*. Furthermore, reduction in CeLEA1 expression by RNAi-mediated gene silencing significantly reduces survival of DJ after several stresses such as desiccation, heat shock, and osmotic shock (40). CbrLEA1, the CeLEA1 homolog in *C. briggsae*, shows high homology to a LEA3-like protein from *P. vanderplanki* ($3e^{-17}$). Two LEA-like EST sequences are described for the peanut pod nematode *Ditylenchus africanus* (39) and one for the meadow nematode *Pratylenchus penetrans*. The ESTs show similarities to LEA-like proteins from *P. vanderplanki*, *Steinernema carpocapsae*, and *C. elegans*, but no experimental evidence on function or expression currently exists. *Steinernema feltiae* and *S. carpocapsae* are entomopathogenic nematodes and can be described as slow-dehydration strategists. Infective juveniles (IJ) of *S. feltiae* accumulated a LEA-like protein approximately 10-fold above control when exposed to 97% relative humidity (RH) for three days (36), and message for another LEA-like protein (SfLEA1) increased more than 100-fold after exposure to 97% RH for 24 h (38). The message for several LEA-like proteins also increased in response to desiccation and osmotic shock in the closely related species *S. carpocapsae* (37). An EST sequence encoding a protein with homology to the nematode LEA-like protein CeLEA1 ($6e^{-4}$) is expressed in the arctic springtail (*Onychiurus arcticus*) (49), and desiccation-induced proteins that cross-react with an antiserum against ArLEA1A have been described for several soil- and surface-dwelling Collembola (48). Three different LEA-like proteins are upregulated in final instar larvae of the anhydrobiotic chironomid *P. vanderplanki* in response to desiccation and osmotic stress (50). PvLEA1 and PvLEA3 exhibit high homologies to the *C. elegans* LEA-like protein CeLEA1 ($9e^{-32}$ and $3e^{-13}$), whereas PvLEA2 exhibits higher homology to a LEA domain–containing protein from *A. thaliana* ($1e^{-5}$).

DJ: dauer juvenile
RH: relative humidity

SUBCELLULAR TARGETING OF LEA PROTEINS

As evident from the above overview of LEA distributions in animals, multiple LEA proteins are found frequently within a single organism. Such a pattern may reflect subcellular targeting of LEA proteins to protect critical cellular components from desiccation-induced damage. For example, maintaining the integrity of mitochondria in the dry state may be critical, especially given the potential release of proapoptotic factors from the intermembrane space or the disruption in oxidative phosphorylation that might otherwise occur upon rehydration (55). Stabilization of the inner mitochondrial membrane and the enzyme systems of the matrix requires that protective molecules be able to pass through the double-membrane system (56). Nucleus-encoded proteins bound for the matrix or inner mitochondrial membrane typically contain a targeting presequence, which is recognized by the highly conserved import and incorporation machinery of the organelle (57, 58). Localization of LEA proteins to both the cytoplasm and subcellular organelles has now been documented in plant and animal species.

In plants, the group 3 LEA protein (PsLEAm) resides in the mitochondrial matrix of seeds and can be induced in leaves of the pea plant *Pisum sativum*. The 358-amino-acid protein is composed largely of hydrophilic and charged residues typical of LEA proteins, with the exception of a hydrophobic presequence found at the N terminus. A fusion protein composed of this leader sequence and green fluorescent protein (GFP) was expressed in pea leaf protoplast. The resulting mitochondrial localization of GFP demonstrated for the first time in plants the existence of a mitochondrial presequence for a LEA protein. To identify in which submitochondrial compartment the

LEA protein was located, Grelet et al. (59) used detergents to differentially solubilize mitochondrial marker proteins for the outer membrane, intermembrane space, inner membrane, and matrix. Cosolubilization of PsLEAm with fumarase established that the LEA protein resides in the matrix.

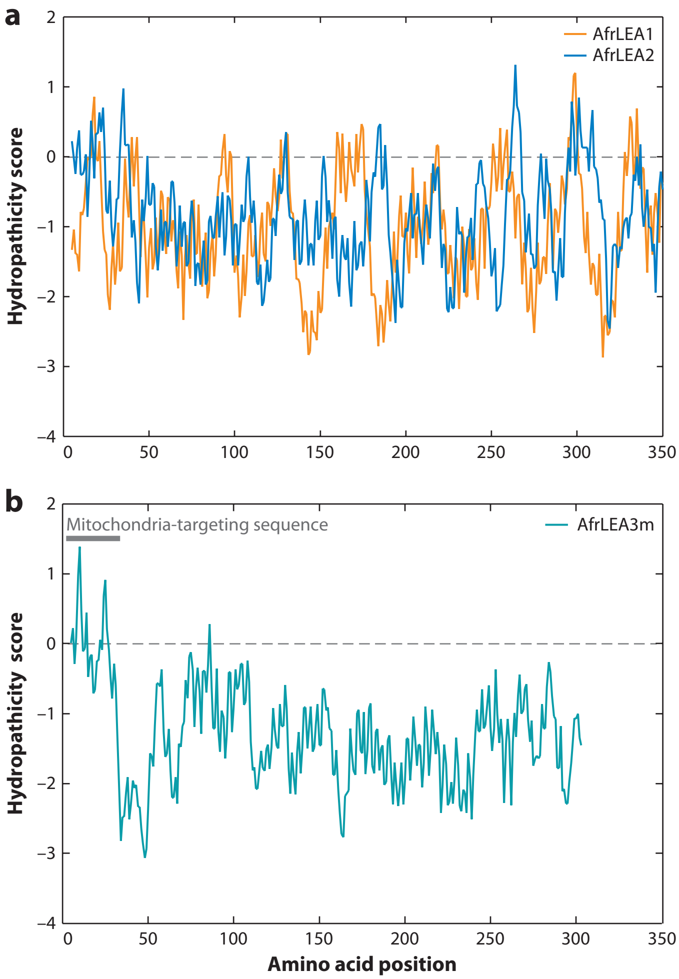

Figure 2

(*a*) Hydropathy plots for the deduced LEA proteins AfrLEA1 and AfrLEA2 indicate strong hydrophilicity, as shown by values less than zero. Modified from Reference 46. (*b*) Hydropathy plot for the deduced protein AfrLEA3 also indicates the strong hydrophilic nature of the protein. The more hydrophobic N-terminal region of approximately 30 amino acids is a result of the mitochondria-targeting sequence. Modified from Reference 44.

The first LEA protein from an animal species reported to be mitochondria targeted was the group 3 AfrLEA3m from *A. franciscana* (44). This group 3 LEA protein is composed of 307 amino acids (**Table 1**) and contains a 29-amino-acid presequence at the N terminus. The hydropathy plot reveals that AfrLEA3m is very hydrophilic, with the exception of the presequence, which is relatively hydrophobic (**Figure 2*b***). Menze et al. (44) showed that when a nucleotide construct encoding the AfrLEA3m presequence was ligated to the nucleotide sequence for GFP and transfected into human hepatoma cells, the chimeric protein was expressed and imported into mitochondria (**Figure 3**). The mitochondrial network was clearly visualized as containing GFP, as verified by colocalization of MitoTracker Red (**Figure 3*b***, frames *1*–*3*). In contrast, the GFP lacking the AfrLEA3m leader sequence was not targeted to mitochondria and did not colocalize with MitoTracker Red (**Figure 3*a***, frames *1*–*3*). The results demonstrated that a mitochondria-targeting sequence is an intrinsic component of AfrLEA3m and strongly suggested that AfrLEA3m is naturally localized to mitochondria of *A. franciscana*. The results also highlight the highly conserved nature of the protein import machinery for mitochondria of mammalian and invertebrate cells and, indirectly, of the targeting sequence (44).

The LEA proteins from the bdelloid rotifer *A. ricciae*, ArLEA1A and ArLEA1B, have a hydrophobic 19-amino-acid N-terminal sequence as well as a putative endoplasmic reticulum retention signal (the amino acid sequence ATEL) at the C terminus. Therefore, both proteins are likely targeted to, or transported through, the endoplasmic reticulum, although this conclusion awaits experimental evidence (53).

In plants, LEA proteins are localized in the cytoplasm, nucleus, mitochondrion, chloroplast, endoplasmic reticulum, vacuole, peroxisome, and plasma membrane (24). An expanded subcellular distribution of LEA proteins in desiccation-tolerant animals may well be revealed in the future.

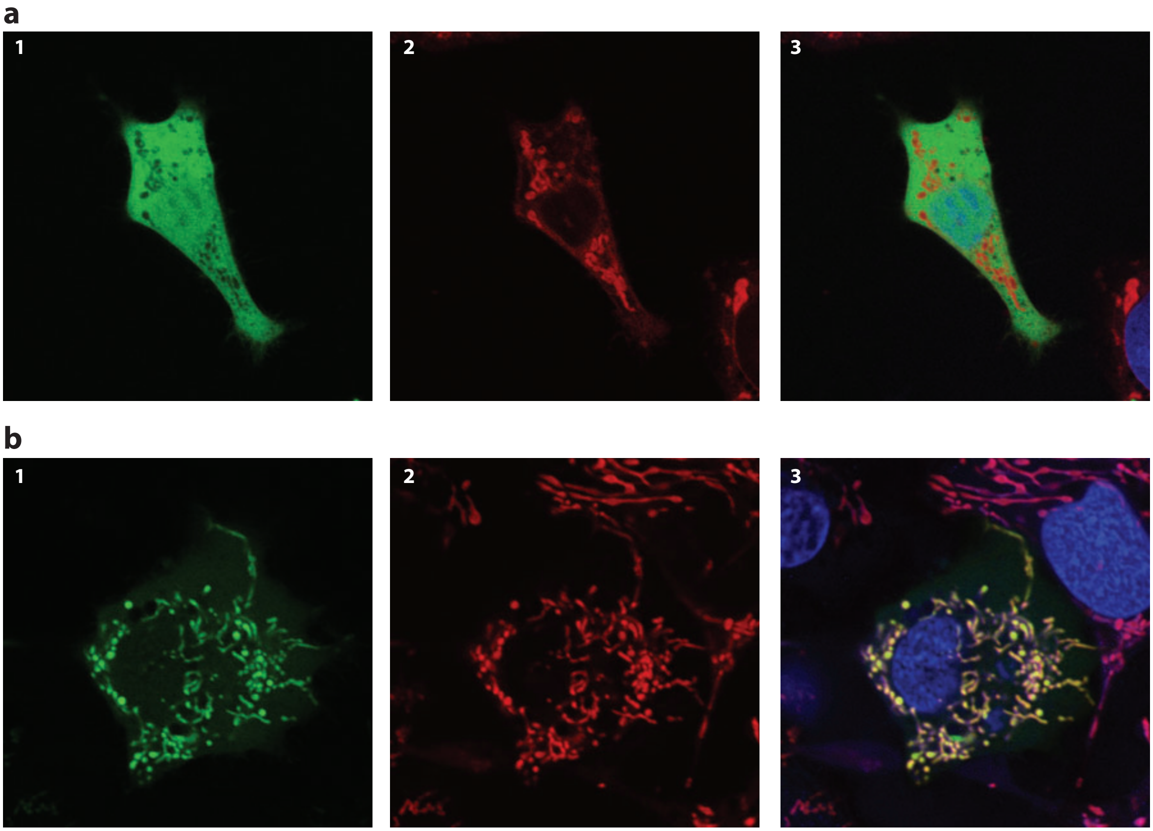

Figure 3

Human hepatoma cells (HepG2/C3A) transfected either with an expression vector encoding a chimeric protein composed of the leader sequence from AfrLEA3m plus green fluorescent protein (GFP) (panel *b*, frames *1–3*) or with a vector encoding only GFP (panel *a*, frames *1–3*). Costaining with MitoTracker Red highlights the mitochondrial network (*red*) (*a*, *2*; *b*, *2*). Green and red fluorescence are colocalized in cells transfected with the chimeric protein (*b*, *3*), but not in cells transfected with GFP only (*a*, *3*) when GFP remains in the cytoplasm. MitoTracker Red fluorescence is not found in the nucleus (*blue*) (*a*, *3*; *b*, *3*). Modified after Reference 44.

STRUCTURE OF LEA PROTEINS IN HYDRATED AND DRY STATES

Hydrophilic Nature and Random Coil in Aqueous Solution

As is clear from the above discussion, a fundamental biochemical feature of LEA proteins constituting the major classification groups is their strong hydrophilic nature. Indeed, traditional Kyte and Doolittle hydropathy plots show that virtually all stretches of sequence for a given LEA protein fall in the hydrophilic space of such a plot. For example, each deduced amino acid sequence for group 3 LEA proteins from *A. franciscana* embryos exhibits negative hydrophobicity scores (**Figure 2*a*,*b***). As discussed above, the exception is the presequence that targets a protein to a subcellular organelle. These leader sequences are more hydrophobic than the mature, cleaved sequence (e.g., References 44 and 59; **Figure 2*b***). In the fully hydrated state, LEA proteins are predominately unstructured with a preponderance of random coil. In aqueous solution, far-UV circular dichroism (CD) spectroscopy did not detect significant secondary structure motifs for the group 3 LEA protein from the nematode *A. avenae* AavLEA1; fluorescence emission spectroscopy confirmed that the single hydrophobic tryptophan residue of AavLEA1 is fully solvent exposed. Indeed, LEA proteins are considered members of the broader classification of intrinsically disordered proteins (e.g., References 18, 19). Their hydrophilic nature fosters an accentuated interaction with solvent water, as discussed further below.

Increase in Secondary Structure During Drying

Of particular relevance to their proposed functions during conditions of dehydration, LEA proteins exhibit the remarkable ability to become more ordered and to develop secondary structure as dehydration proceeds. For animal LEA proteins, Goyal et al. (60) first demonstrated this phenomenon by using Fourier-transform infrared (FTIR) spectroscopy. FTIR spectroscopy allows for the assessment of protein secondary structure in the dry state by using the profile of the amide-I band, which provides information on the relative contributions of α-helix, β-sheet, and turn structures (61). AavLEA1 showed gain of structure during dehydration that was fully reversible upon rehydration. Wolkers et al. (61) originally discovered this pattern for a group 3 LEA protein from *Typha latifolia* pollen. The *T. latifolia* protein in solution was largely in a random-coil conformation, but it was primarily α helical after fast drying. Interestingly, slow drying reversibly led to both α-helical and intermolecular extended β-sheet structures, which suggested that the final protein conformation was not predetermined. Increases in secondary structure during drying have also been reported for LEA proteins from groups 1 and 2 (dehydrins) (62, 63) and for other group 3 proteins and peptides (64, 65).

A recent modeling study by Li & He (66) utilized a 66-amino-acid fragment of AavLEA1 and documented, through molecular dynamics simulation, many of these properties. The simulation of Li & He (66) used one molecule of the AavLEA1 fragment and various numbers of water molecules. These ratios mimicked aqueous solutions of the LEA protein at different water contents from 83.5 wt% to 2.4 wt%. As water is removed, the protein assumes a more folded conformation. At 83.5% water, the LEA protein is fully solvated with water molecules until the water is decreased to 50%. At 50% water, 422 molecules of water are predicted to interact with each molecule of LEA protein (**Figure 4**). In this range of between 83.5% and 50.4%, the protein is unstructured. Below this point, water molecules no longer are sufficient to fully solvate the protein. At less than approximately 20% water (105 water molecules per protein), the protein becomes more dehydrated and begins to adopt a significant amount of secondary structure. α-Helical structure is apparent, and hairpin-like structures form (**Figure 4**). At 2.4% water (10 water molecules per protein), the structure is very similar to that in the complete absence of water (66). These structural changes are observed during dehydration of the native LEA proteins (60, 64), and Li & He (66) suggest that because these compact, hairpin structures appear only at very low water percentages, a functional role for the protein is in the dry state rather than in the hydrated state. **Figure 5** shows the formation of various secondary structures (i.e., α helix, random coils, and β sheet) as a function of water removal. Secondary structure is predominantly random coil at water content of more than 20%, again matching experimental results with CD and FTIR (e.g., References 60, 61, 64, and 65). In the dehydrated state, the structure is primarily α helical.

Structural Stability and Intraprotein Hydrogen Bonding Increases During Desiccation

Consistent with the gain of secondary structure as LEA proteins are dehydrated, the numbers of intraprotein hydrogen bonds are projected to increase and the number of hydrogen bonds present between the protein and water are projected to markedly decrease as water is removed (**Figure 6*a***). Both projected changes occur primarily at 20% water and below.

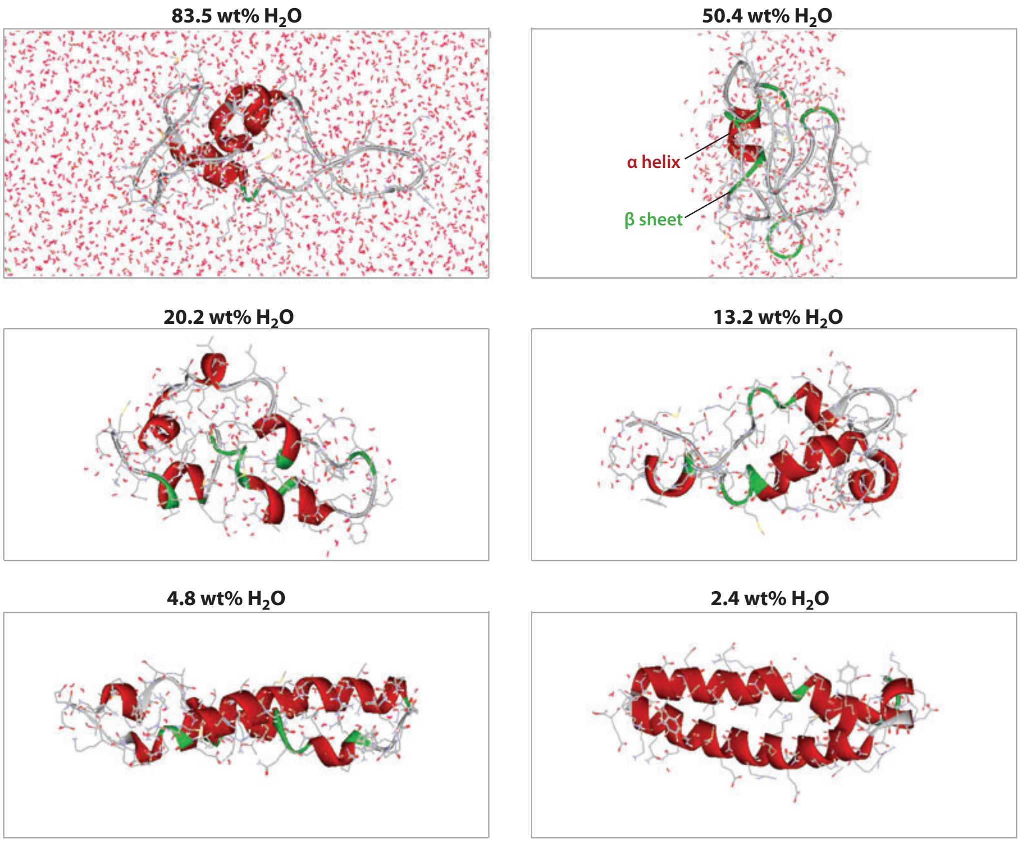

Figure 4

Representative conformations of the 66-amino-acid fragment of a LEA protein (AavLEA1) from the nematode *Aphelenchus avenae* are shown at different water contents. The smaller water molecules (*gray* and *red*) are depicted in the line style, and the larger LEA protein molecules are denoted using the solid ribbon style (α helix, *red*; β sheet, *green*; random coil, *gray*). Modified from Reference 66.

Estimates of overall protein stability show that the LEA protein is much more stable at low water percentages, as judged by RMSD (root-mean-square deviation) of all atoms on the protein backbone and MDTF (mean dihedral transition frequency) of all amino acids in the protein (**Figure 6*b***). The smaller the two parameters, the more stable is the structure of the protein. RMSD and MDTF change dramatically below 20% water and suggest a strongly stabilized structure at <5%. Li & He (66) suggest that the structural flexibility of the LEA protein in the aqueous condition, versus the compact, folded three-dimensional shape when water is limiting to macromolecular hydration, may contribute to the respective chaperone and "molecular shield" functions that have been proposed, and to various degrees supported, by experimentation (discussed below).

PHYSIOLOGICAL ROLES OF LEA PROTEINS WITH AND WITHOUT SUGARS

Animals with natural desiccation tolerance often accumulate low-molecular-weight organic solutes (e.g., trehalose; sucrose is common in plants) along with protective macromolecules like LEA proteins. Such dual protection with protein and sugar is not always the case, as bdelloid rotifers (43, 67) and one species of tardigrade (68) that do not accumulate

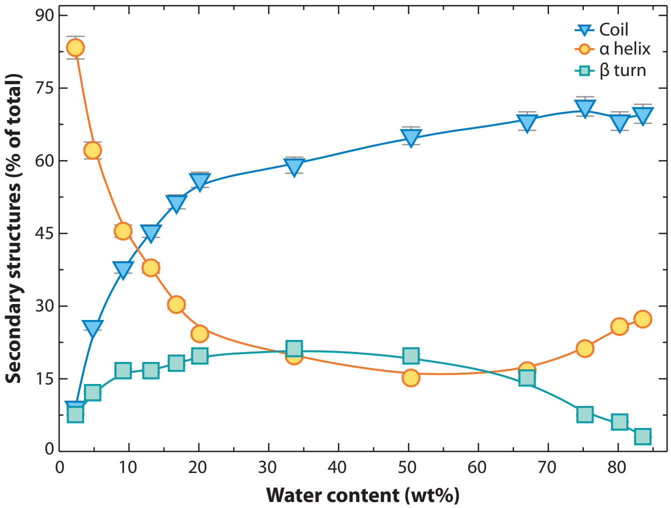

Figure 5

Percentages of the three major secondary structures in the LEA protein fragment from the nematode *Aphelenchus avenae* as a function of water content. Symbols represent data from a molecular dynamics (MD) simulation, and the lines are B-spline fits to the MD data. Each error bar represents ± one standard deviation. Redrawn from Reference 66.

protective sugars exemplify. Nevertheless, trehalose is clearly synthesized during the acquisition of desiccation tolerance in nematodes (69, 70) and other organisms (71). Crowe and colleagues (69, 72) reported the compelling observation that unless time is sufficient for *A. avenae* to convert glycogen to trehalose prior to desiccation, the nematode does not survive drying; if conversion to trehalose is accomplished during slow drying, the organism survives dehydration. Apparently trehalose buildup may not be sufficient for anhydrobiosis (26). As Hoekstra and colleagues (26) emphasized, for plant desiccation tolerance, one specific mechanism for protection does not confer tolerance on its own; the interplay of several mechanisms is essential.

Protection of Proteins and Organelles

LEA proteins have the capacity to protect target proteins from inactivation and aggregation during water stress, including both freezing and drying. Protection against freezing damage has been extensively studied for group 2 LEA proteins (for a review, see Reference 24) and has been reported for group 3 members (73, 74). Reyes et al. (75, 76) showed that group 2 LEA proteins can protect lactate dehydrogenase against both drying and freezing, and because the details of the enzyme inactivation appear to be different, these LEA proteins may be capable of protecting against multiple forms of structural alteration. Protection against damage during drying is widespread among various groups of LEA proteins from plants and animals (59, 74, 77, 78).

Although the group 3 LEA protein from *A. avenae* (AavLEA1) can protect citrate synthase against desiccation-induced aggregation in the absence of any other desiccation protectant like a sugar (53, 74), it also displays a synergistic stabilization in the presence of trehalose. There is precedence for such a synergistic protection of proteins against heat stress and during the facilitation of protein folding by trehalose and non-LEA proteins (79–81). Investigators reported a similar observation with mammalian cell lines for trehalose and the small stress protein p26 (82).

Goyal et al. (74, 77) proposed that LEA proteins, at least in some cases, exert their protein antiaggregation function during water removal by sterically preventing interactions between partially unfolded proteins that would otherwise form intracellular aggregates, i.e., molecular shielding. However, because desiccation of globular proteins often induces aggregation and denaturation (18), LEA proteins would be ineffective at preserving the functionality of target proteins in many cases unless LEA proteins also serve as chaperones; i.e., simply preventing aggregation would be insufficient. Whether LEA proteins also form specific complexes with clients, as do chaperone proteins, is difficult to demonstrate experimentally during drying because the proteins become forced together (cf. Reference 24).

AavLEA1, a LEA protein found in nematodes, maintains this antiaggregation capability in vivo (78). Researchers developed several human cell lines in which the expression of AavLEA1 could be induced. These cells were then tested for antiaggregation activity with a model protein, EGFP-HDQ74, which is prone to spontaneous aggregation. Cells induced to express AavLEA1 immediately after transfection with EGFP-HDQ74 exhibited significantly less aggregation than cells that did not express AavLEA1.

Finally, evidence suggests that trehalose and LEA protein stabilize mitochondria during freezing (44). Mitochondria isolated from *A. franciscana* embryos contain trehalose inside the matrix naturally, and at least one LEA protein (AfrLEA3m) is targeted to the mitochondrial matrix (see above). When these mitochondria are frozen in a trehalose solution (with trehalose and AfrLEA3m naturally present internally), remarkably high respiratory control ratios (succinate plus rotenone) for frozen/thawed mitochondria were observed compared with ratios for control, nonfrozen mitochondria. These respiratory control ratio values are much higher than those reported for mammalian mitochondria frozen/thawed without LEA protein and trehalose present only outside (83).

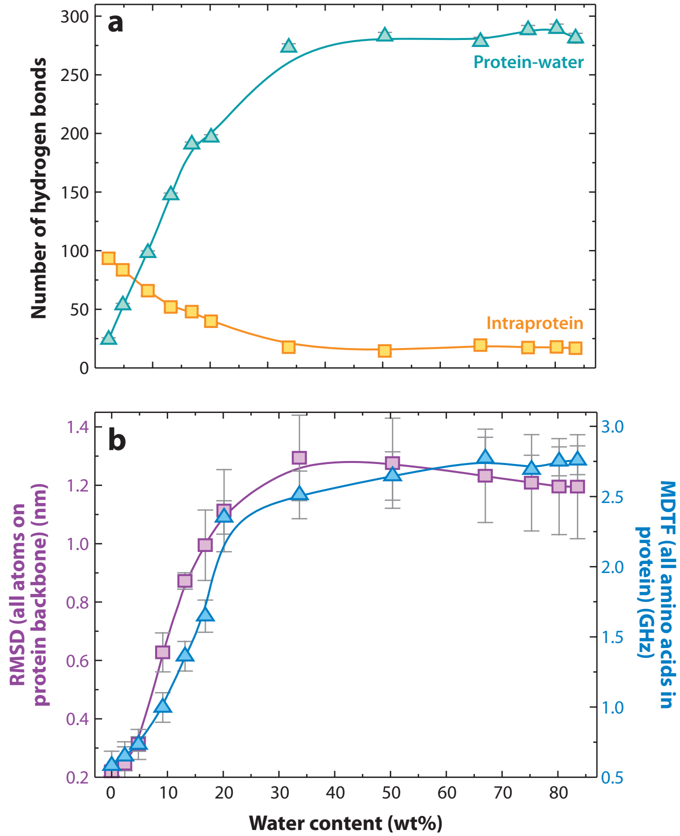

Figure 6

(*a*) Number of intraprotein and protein-water hydrogen bonds as a function of water content. (*b*) RMSD (root-mean-square deviation) of all atoms on the protein backbone and MDTF (mean dihedral transition frequency) of all amino acids in the protein as a function of water content. In both panels *a* and *b*, symbols represent data from the molecular dynamics (MD) simulation, and the lines are B-spline fits to the MD data. Each error bar represents ± one standard deviation. Redrawn from Reference 66.

Interactions with Membranes

The first evidence that LEA proteins interact with membranes came from work with COR15a from *A. thaliana* (84). COR15a seems able to protect membranes from undergoing detrimental phase transitions during freezing (84). Although not classified as a LEA protein at the time, Wise (54) later identified COR15a as a group 3 LEA protein. Membrane interaction has since been shown for many other plant LEA proteins, but the function of these interactions is not always clear (for a review, see Reference 24).

Molecular modeling of the amphipathic α helices that form upon dehydration of a LEA protein from pea seed mitochondria (PsLEAm) reveals patterns that resemble class A amphipathic helices of apolipoproteins (64). Nearly all the negative amino acid residues in PsLEAm form a stripe that is bordered on either side by positive residue stripes. Tolleter et al. (64) suggested that this pattern indicates the ability of PsLEAm to interact with phospholipid membranes in the dry state because PsLEAm assumes this α-helical structure only upon desiccation and then is able to integrate into the membrane parallel to its plane (64). Furthermore, this group showed that PsLEAm protects liposomes during drying by preventing membrane fusion events (64) and lysis (85). Differential scanning calorimetry (64) and FTIR spectroscopy (85) also indicated interaction of PsLEAm with liposomes in the dry state.

Although most of the evidence for membrane interaction to date involves LEA proteins from plants, this characteristic has also been

Amphipathic: containing both polar (hydrophilic) and nonpolar (hydrophobic) regions

T_m: melting temperature, or the temperature at which a gel-to-liquid crystalline phase transition occurs

T_g: glass-transition temperature

Vitrification: the transition of a substance into an amorphous glass state, in which rates of molecular diffusion are greatly reduced

reported in animals. Pouchkina-Stantcheva et al. (53) used FTIR to show that a LEA protein from a bdelloid rotifer (ArLEA1B) interacts with dried liposomes. ArLEA1B is a group 3 LEA protein but has atypical characteristics, as discussed above. Unlike typical group 3 LEA proteins that are unstructured in solution, ArLEA1B has an α-helical structure and is not able to act as a molecular shield (53). In fact, the opposite result occurs. When CS is dried in the presence of ArLEA1B, there is increased aggregation compared with CS dried alone. However, ArLEA1B interacts with dry phospholipid membranes to a greater extent than do ArLEA1A and AavLEA1, which are able to protect CS. The presence of ArLEA1B significantly decreases the gel-to-liquid crystalline phase-transition temperature (T_m) of dry liposomes (53).

That individual LEA proteins may have different protective abilities is a possible additional explanation for why there are multiple LEA proteins in a single organism. Multiple LEAs may be required, not only for targeting to different organelles but also to protect different cellular components (e.g., proteins versus membranes) at a given location in the cell. Both ArLEA1A and ArLEA1B contain a C-terminal endoplasmic reticulum retention signal and an N-terminal hydrophobic region, indicating that they are targeted to the same region in the cell, but the two proteins seem to have very different functions (53).

Stabilization of Sugar Glasses

Wolkers et al. (61) initially reported that LEA plant proteins can stabilize vitrified sugar glasses (trehalose, sucrose), as judged by shifts in the glass-transition temperature (T_g). Vitrification is thought to be one important property for substantial dehydration tolerance (cf. References 71 and 86). As described by Hoekstra et al. (26), a sugar glass is "an amorphous metastable state that resembles a solid, brittle material, but with retention of the disorder and physical properties of a liquid. In a glass state, the rates of molecular diffusion and chemical reactions are greatly reduced." Using air-dried sugar or sugar-protein mixtures, Wolkers et al. (61) monitored glass melting during sample heating with FTIR by observing the position of OH stretching vibration of the sugar, and they estimated T_g values from frequency-versus-temperature plots. Compared with the T_g for pure sucrose (60°C), the addition of 0.5 mg of LEA protein per milligram of sugar raised the T_g to 69°C, and the addition of 0.8 mg of protein per milligram of sucrose further raised the T_g to 79°C. High T_g values that are considerably greater than ambient temperature are considered physiologically beneficial for an organism so that the vitrified state is preserved (i.e., no melting of the sugar glasses occurs) as the environmental temperature increases. However, the ability of LEA protein to increase the T_g of sugar glasses is not unique but is also seen, for example, with poly-L-lysine (87) and hydroxyethyl starch (88).

More recently, a similar impact on T_g was documented for model synthetic peptides composed of two or four 11-mer motifs of group 3 LEA proteins from insects, nematodes, and plants (65). Thus, each peptide contained 22- or 44-amino-acid residues. A control peptide (with identical amino acid composition but with randomized sequence) was used for comparison. Mixtures of trehalose and experimental LEA-like peptides exhibited a 7°C increase in T_g in peptide-trehalose mixtures compared with mixtures using the control peptide. Unfortunately, a direct experimental comparison to pure trehalose was not possible due to lower water content in the pure-trehalose samples versus the peptide mixtures; the presence of even small amounts of water after drying lowered the T_g of the sugar mixtures (cf. Reference 89). In addition, a larger shift toward lower wave numbers was seen with the experimental peptide mixtures versus control peptide mixtures, which indicates that the hydrogen bonding network in the glassy state was strengthened with the LEA-like peptide compared with the control peptide (65). The work of Wolkers et al. (61) and that of Shimizu et al. (65) support the general concept that the presence of LEA proteins stabilizes sugar glasses (e.g., Reference 60). A T_g elevated above the T_g of a pure sugar as reported by Wolkers et al. (61) may be of physiological importance when environmental temperatures rise and/or cellular water content increases in desiccated animals. In the future it would be helpful to relate the ratios of sugar:LEA protein used for in vitro studies to those existing in vivo in cells in order to support the physiological relevance of these findings.

The synthetic peptides used by Shimizu et al. (65) vitrified at a high T_g (>100°C); the control (randomized) peptide also exhibited a fairly high T_g, although the T_g was approximately 18°C lower than that of experimental peptides. Shimizu et al. (65) suggested that the dried LEA proteins alone should act as good desiccation protectants, but such a possibility would need to be verified with native LEA proteins.

Hydration Buffers and Ion Sequestration

Another suggested function for LEA proteins is as a "hydration buffer" whereby unstructured hydrophilic proteins bind greater numbers of water molecules in their hydration shells than does a typical globular protein (e.g., References 63 and 90–92). Bokor et al. (93) estimated with nuclear magnetic resonance (NMR) that several intrinsically unstructured proteins (including *A. thaliana* ERD10, a group 2 LEA) bound 20–50% more water compared with bovine serum albumin. Thus, theoretically LEA proteins may retard water loss during dehydration, but does this water binding represent a significant fraction of the total water content in a cell? Additional direct measurements of the hydration water for LEA proteins with various NMR approaches and thermodynamic-linkage studies under osmotic stress (94–96) are needed to answer this question.

In cotton seed, the LEA protein D-113 approaches 4% of the total cytosolic protein (91). This fraction represents a LEA concentration in the aqueous phase of undesiccated seeds of 283 μM (91) or approximately 2.11×10^8 LEA molecules in a 15-μm-diameter cell with 70% water content by our calculations. From the water binding capacity modeled for a 66-amino-acid fragment of the group 3 nematode LEA protein (66) and adjusting for differences in molecular mass, we calculate that the plant LEA protein can bind approximately 4,140 water molecules per LEA molecule or 8.74×10^{11} water molecules per cell. In the water volume calculated for a single cell are 4.14×10^{13} water molecules. Thus, these calculations suggest that this LEA protein organizes roughly 2% of total cellular water. A second LEA protein (D-7) expressed in cotton seed represents 2.6% of cytosolic protein, which in combination with D-113 could raise the organized water attributable to LEA proteins in cotton seed to more than 3%. Nevertheless, LEA-specific binding would influence the water-loss kinetics of only 2–3% of cell water. It seems unlikely that the release of this small water fraction during desiccation would be of significant biological advantage, particularly given the many different drying regimes and rates in nature and the restricted range of water content across which such release would occur (cf. **Figures 4–6**).

Hydration buffer: the concept that unstructured proteins bind large numbers of water molecules, which can be released inside cells during dehydration to provide beneficial protection

To our knowledge, there are no direct measurements of changes in the kinetics of cellular water loss attributable to the presence of LEA proteins. Over days of development within the seed capsule of intact *A. thaliana*, mutant seeds deficient in expression of ATEM6 (group 1 LEA protein) reached a dehydrated state earlier than wild-type seeds on the basis of morphology (97). However, additional developmental characteristics also appeared to be accelerated in the mutant. Thus, conclusions about absolute differences in water-loss rate between mutant and control seeds could benefit from the measurement of water kinetics in isolated seeds. For animals, other strategies for slowing the rate of water loss would likely be more quantitatively significant. Such strategies include anterior-posterior contraction in tardigrades [i.e., formation of the "tun" or *tonnchenstadien* (98–101)], coiling in nematodes (e.g., References 102 and 103), morphological changes in cuticular lipids [the "permeability slump" (104, 105)], and alterations in the egg envelope and perivitelline space that reduced water permeability in diapause fish embryos (106).

Another function often proposed for LEA proteins, due primarily to their hydrophilic and charged nature, is ion sequestration. As cells dehydrate, the overall concentrations in inorganic ions in the cell rise and potentially disrupt enzyme function (cf. Reference 107). However, because an animal cell contains 130–150-mM K^+ (in addition to the anion load) and K^+ levels increase during dehydration, it is unlikely that general ion binding by LEA proteins significantly retards free-ion increase during dehydration. Even though the number of amino acid residues with ion chelating function is high and the percentage of total cell protein represented by LEA proteins is often substantial, the number of chelation sites is minor relative to the number of total free ions. Furthermore, when LEA proteins release their hydration water during desiccation (**Figure 4**), the ions bound to the proteins typically follow suit and are also released, thereby voiding any benefit, unless there are highly specific binding sites for ions like Ca^{2+} (see below). Thus, in our view general ion sequestration is transient and inconsequential during cell desiccation.

There may be a more physiologically relevant impact of ion sequestration for divalent ions, including calcium and metal ions like Fe^{3+}, Cu^{2+}, and Zn^{2+}, when one considers their nanomolar concentrations in cells. Although these binding activities have not been studied for LEA proteins of animals, data support specific Ca^{2+} binding sites for plant homologs. Furthermore, there are reports that group 2 (dehydrin) proteins show antioxidant properties indirectly by sequestering metal ions and directly by scavenging reactive oxygen species (for a review, see Reference 24).

CONCLUDING REMARKS AND FUTURE ISSUES

Understanding the mechanisms that organisms have evolved to tolerate drying and freezing will inform us about fundamental ways by which cells and tissues can survive water limitation. The roles of LEA proteins in this context are only beginning to emerge decades after their discovery. More work in vivo is needed to support the important inferences from many excellent in vitro studies.

Among the key biological issues still requiring complete explanation is the evolutionary origin of these proteins, once thought to exist only in land plants. Discoveries of LEA expression in bacteria, fungi, and multiple animal phyla suggest a primitive ancestral origin of LEA proteins coupled to subsequent loss of expression in numerous lineages. The consistent expression of LEA proteins under conditions of water stress in animals and other organisms supports their role in conferring desiccation tolerance, but definitive mechanisms for this presumed function are incompletely known (cf. Reference 24). For example, do LEA proteins confer protection to macromolecules by acting as chaperones, molecular shields, or both? Perhaps their action depends on the nature of the specific perturbation involved or whether the LEA proteins are working in concert with sugars or other protective solutes. Why is the presence of LEA proteins without protective sugars sufficient for desiccation tolerance in some anhydrobiotic species, but in other tolerant animals, high concentrations of trehalose are preferentially accumulated during drying together with LEA proteins? Maybe the absolute cellular titer of LEA proteins attainable governs the apparent need for trehalose. More quantitative measurements of the hydration water of LEA proteins and direct measurements of differential rates of water loss in the presence and absence of LEA proteins will help clarify whether the concept of hydration buffers is realistic.

More information on the subcellular compartmentalization of LEA proteins, particularly in animal cells, is required; plant studies are much further ahead in this regard. Quantitative measurements of protein expression levels in cells and subcellular compartments in animals would help guide the design of in vitro stabilization studies that would more closely reflect physiologically relevant ratios of protein to sugar. Such information would be equally important during ectopic expression of LEA proteins in cells for the purpose of conferring desiccation tolerance for applied, biomedical purposes (cf. Reference 56).

SUMMARY POINTS

1. Exciting studies now show that LEA proteins are a component in the arsenal of defense mechanisms utilized by many animals that survive severe water stress. Suggested roles of LEA proteins in animals include the protection of proteins through chaperone activity and/or molecular shielding, interaction with and stabilization of phospholipid membranes, and reinforcement of vitrified sugar glasses.
2. Most LEA proteins are highly hydrophilic and intrinsically unstructured in aqueous solution. One extraordinary feature of LEA proteins is their ability to increase secondary (predominantly α-helix) structure during drying. This feature suggests important functional differences for LEA proteins in the dried state. As folding of LEA proteins occurs during water removal, the number of intraprotein hydrogen bonds increases, and the number of hydrogen bonds between the protein and solvent decreases. Estimates of overall protein stability show that LEA proteins are more stable at low water percentages and display greater structural flexibility in aqueous conditions.
3. Although LEA proteins can protect other macromolecules and biological structures, they also act synergistically with nonreducing disaccharides (e.g., trehalose and sucrose) for this purpose. For example, trehalose strongly enhances in vitro stabilization of enzyme activity by LEA proteins during drying. Furthermore, direct evidence shows that LEA proteins increase the glass-transition temperature (T_g) of sugars. Higher T_g values may be physiologically important to dehydrated organisms because the vitrified state is preserved at higher environmental temperatures.
4. Many organisms express multiple isoforms of LEA proteins. The occurrence of more than one type of LEA protein in a single organism suggests multiple subcellular locations and the ability to perform divergent functions. For example, a mitochondria-targeted LEA protein has been documented in a crustacean, and two similar LEA proteins from the same nematode species perform surprisingly different functions.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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